

Monthly Progress Report – Month 3 (2026)

AI-Based Educational Content Viewer with Topic Summarization

Krishna Kanta Handiqui State Open University (KKHSOU)

MCA 4th Semester – Industrial Project

Project Overview

Project Title: AI-Based Educational Content Viewer with Topic Summarization

This project is being carried out at Krishna Kanta Handiqui State Open University (KKHSOU) as part of the MCA 4th Semester Industrial Project. The project aims to develop an intelligent educational content viewer that enables students to read PDF-based study materials through an interactive digital interface and receive AI-generated summaries and learning assistance for improved understanding.

The third month of the project focused on refining and optimizing the features developed in the previous months, improving the accuracy of AI-generated outputs, performing system testing with real educational PDFs, and preparing the project for final demonstration and evaluation.

Objectives of the Project

- Develop an interactive and user-friendly PDF viewer for academic materials.
- Implement page-wise and topic-wise AI-generated summaries.
- Improve the accessibility and readability of educational PDF content.
- Enable users to ask questions directly from the book content.
- Enhance digital learning through intelligent features such as summaries, topic detection, and video recommendations.
- Optimize system performance and conduct comprehensive testing.

Technologies Used

- Python
- Flask (Web Framework)
- PyMuPDF for PDF text extraction
- HTML, CSS, JavaScript
- PDF.js for rendering PDF pages
- Transformer / NLP-based summarization techniques
- JSON-based structured summary storage
- pytest for backend testing

Organization Profile

Krishna Kanta Handiqui State Open University (KKHSOU) is a state university in Assam that provides higher education through open and distance learning. The university emphasizes digital learning solutions and accessible educational resources for students. This project supports that vision by creating an AI-enhanced educational reading platform.

Work Done in Month 3

- Improved the accuracy of AI-generated summaries by refining the NLP summarization pipeline.
- Enhanced the Ask This Book feature to produce more relevant and contextually accurate answers.
- Improved the topic detection algorithm to better identify subjects from the currently displayed page.
- Refined the video recommendation module to suggest higher-quality and more relevant educational videos.
- Optimized page rendering speed and reduced loading time for large PDF files.
- Performed comprehensive testing of the system using real KKHSOU educational PDF materials.
- Fixed bugs identified during testing related to page text extraction and summary matching.
- Improved confidence scoring logic for AI-generated answers.
- Optimized JSON summary storage for faster read access during runtime.
- Completed mobile UI optimization for small screen devices.
- Conducted user interface review to ensure consistent and professional appearance.
- Prepared project documentation and final report draft.
- Organized and cleaned the project codebase for evaluation submission.

Technical Development Details

1. AI Summarization Accuracy Improvements

Significant effort was invested in improving the quality of AI-generated summaries and answers. The summarization pipeline was updated to handle longer and more complex academic texts more effectively.

Improvements Made:

- Refined text pre-processing steps to remove noise from extracted PDF content.
- Improved chunking strategy for long documents to preserve semantic context.
- Updated prompt logic for transformer-based model to generate more concise summaries.
- Validated summaries against original content for quality assurance.

2. Ask This Book – Enhanced Accuracy

The question-answering feature was further improved to deliver more precise and contextually relevant answers based on book content.

Enhancements:

- Improved keyword extraction for better content matching.
- Refined confidence scoring to reflect answer quality more accurately.
- Added fallback logic to handle questions with limited matching content.
- Improved suggested follow-up questions based on the current topic.

3. System Testing

Testing was conducted using real educational PDF documents from KKHSOU to evaluate the system under realistic conditions.

Testing Areas:

- PDF rendering accuracy across different document types and sizes.
- Correctness and relevance of AI-generated summaries and answers.
- Topic detection and video recommendation quality.
- Performance and loading speed for large multi-page PDFs.
- Mobile responsiveness and usability on small screens.
- Feature functionality testing including bookmarks, highlights, and notes.

Challenges Faced

1. Summarization Quality on Dense Academic Text

AI summarization models sometimes produced less accurate summaries for highly technical and dense academic content.

Resolution: Text chunking and pre-processing were improved, and prompts were refined to guide the model toward more focused summaries.

2. Performance with Large PDF Files

Loading and rendering large PDF files with many pages caused noticeable delays in the viewer.

Resolution: Lazy loading was implemented so only the current and adjacent pages are rendered at any given time, significantly reducing load time.

3. Ensuring Consistency Across Browsers

Some UI elements and PDF rendering behaviors differed slightly across different web browsers.

Resolution: Cross-browser testing was conducted and CSS and JavaScript adjustments were made to ensure consistent behavior.

Month 3 Milestones

Milestone	Status	Completion
AI Summarization Accuracy Improvement	Completed	100%
System Testing with Real PDFs	Completed	100%
Mobile UI Optimization	Completed	100%
Performance Optimization	Completed	95%
Project Documentation	In Progress	85%
Overall System Completion	In Progress	92%

Learning Outcomes in Month 3

- Deepened understanding of NLP summarization techniques and their practical limitations.
- Hands-on experience in software testing methodologies for web applications.
- Improved skills in JavaScript for implementing advanced UI features such as highlighting and annotations.
- Better understanding of performance optimization strategies for frontend applications.
- Practical experience in preparing project documentation for academic evaluation.
- Improved knowledge of cross-browser compatibility testing and resolution techniques.

Next Month Plan

- Finalize and complete all project documentation including technical and user manuals.
- Conduct a final round of testing and bug fixing.
- Prepare the project for final demonstration to faculty and evaluators.
- Create a presentation summarizing the project work, features, and outcomes.
- Submit the complete project report as required by the university.

Conclusion

The third month of the project was focused on quality improvement, feature additions, and system testing. Significant progress was made in enhancing the accuracy of AI-generated outputs, and testing the system with real educational content.

The system has now reached a high level of completeness with most planned features implemented and tested. The overall project is on track for final submission and demonstration. The work carried out in Month 3 has transformed the prototype developed in earlier months into a near-complete, functional intelligent educational content viewer ready for evaluation.

Bibliography

- Official Flask Documentation
- Official PDF.js Documentation
- Python documentation for PyMuPDF
- Online technical resources related to AI summarization and NLP
- Web development references for HTML, CSS, and JavaScript
- KKHSOU academic PDF materials used for system testing
- Research references on educational technology and intelligent tutoring systems